CCC-C-421b
May 12, 1967
SUPERSEDING
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November 13, 1963

FEDERAL SPECIFICATION

CLOTH, DENIM, COTTON, SHRUNK AND UNSHRUNK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cloth, denim, cotton, shrunk and unshrunk

1.2 Classification.

1.2.1 Types, classes, and styles. The cloth shall be of the following types, classes, and styles (see table I, 6.2):

Type I—8.04 to 8.78 ounces.
Type II—8.95 to 9.75 ounces.
Type III—9.85 to 10.97 ounces.
Type IV—11.06 to 12.30 ounces.

Class 1—Unshrunk.
Class 2—Shrunk.

Style A—Whiteback.
Style B—Colored or tinted.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CCC-T-191—Textile Test Methods.
PPP-P-1134—Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics).

Federal Standard:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulation Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C., 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the cloth shall match the standard sample for shade and shall be equal to, or better than, the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material. The cotton shall be thoroughly cleaned, carded, drawn, and evenly spun into single or ply yarns for both the warp and filling.

3.3 Color. The color shall be as specified and shall match the standard sample when available (see 3.1, 6.2, and 6.3). Style A, whiteback cloth shall contain dyed warp yarns and white filling yarns. When Blue 3301 shade is specified (see 6.2 and 6.4), the shade shall be obtained by the use of vat dyestuffs. The filling yarns may be either blended or tinted. Style B cloth shall contain dyed warp and dyed or tinted filling yarns.

3.3.1 Matching. The dyed cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 Colorfastness. Unless otherwise specified (see 6.2), the dyed cloth shall show colorfastness to light (after 40 hours' exposure) and laundering, equal to, or better than, the standard sample when tested as specified in 4.4. When no standard sample is available, or when the standard sample is not a reference for colorfastness, the dyed cloth shall show good colorfastness to light (after 40 hours' exposure), laundering, and dry crocking. In addition, the dyed cloth shall show fair colorfastness to wet crocking when tested as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the minimum requirements specified in table I when tested as specified in 4.4.

3.5 Weave. Unless otherwise specified, the weave shall be a 3-harness construction. $\frac{2}{1}$ right-hand twill (see 6.2).

3.5.1 Width. The width exclusive of selvages shall be as specified (see 6.2).

TABLE I. Physical requirements

Type	Class	Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength, grab method (minimum)	
			Warp	Filling	Warp	Filling
		Ounces			Pounds	Pounds
I	1	8.04	62	39	145	53
	2	8.78	63	42	145	58
II	1	8.95	66	43	150	65
	2	9.75	67	46	150	70
III	1	9.85	68	42	167	70
	2	10.97	69	45	167	75
IV	1	11.06	70	42	177	85
	2	12.30	71	45	177	90

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3.6 Nonfibrous materials. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed the percentage for the applicable type as shown below when tested as specified in 4.4.

Type	Percent
I	14
II, III, and IV	13

3.7 Shrinkage or stretch. Class 2 cloth shall be preshrunk and shall not have a residual shrinkage or stretch of more than 2.0 percent in either direction of the warp or filling when tested as specified in 4.4. The preshrinking process used shall not be identified by name or trademark, either on the cloth or the ticket or package.

3.8 pH. The pH of the cloth shall be not lower than 7.0 nor more than 10.0 when tested as specified in 4.4.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be in continuous lengths, each not less than 40 yards. Each length shall be put up in flat folded pieces as specified in PPP-P-1134.

3.10 Fiber identification. Each piece shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulation Under the Textile Fiber Products Identification Act.

3.11 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examination of the end item. Examination of the end item shall be in accordance with provisions of MIL-STD-105 and 4.2.2 through 4.2.5.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.2.2 Yard-by-yard examination. Thirty yards from each piece in the sample shall be examined on the face side. The sample shall consist of 25 pieces taken from 25 containers. All defects as defined in section I of Fed. Std. No. 4, which are clearly noticeable at normal inspection distance (3 feet), shall be scored and assigned demerit points as listed in 4.2.2.1, except that only those knots and slubs which exceed the limits shown on figure 1 shall be scored. No one of the 30 continuous linear yards (folded yards) or increments of one yard on the measuring device of the inspection machine selected for examination from any one piece within the sample size shall be penalized more than four points. The lot shall be unacceptable if the points per hundred square yards exceed 40.00. Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3{,}600}{\text{Contracted width of cloth (inches)} \times 750} = \text{Points per 100 square yards.}$$

4.2.2.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension—one point.
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension—two points.

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For defects exceeding 6 inches, but not exceeding 9 inches in any dimension—three points.
For defects exceeding 9 inches in any dimension—four points.

4.2.3 Examination for length.

4.2.3.1 Individual pieces. During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

4.2.4 Examination for shade. During the yard-by-yard examination, each piece in the sample shall be examined for shade. Any piece in the sample off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.5 Examination for identification of preshrinkage process and noncompliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket or are not labeled, ticketed, or invoiced in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP–P–1134 to determine that packaging, packing, and marking requirements of section 5 are complied with.

4.4 Testing of the end item. The methods of testing specified in CCC–T–191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3-½ continuous yards, full width of cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The lot size shall

TABLE II. Test methods

Characteristic	Requirement reference	Test Method No.
Colorfastness to:		
Light	3.3.2	5660
Laundering	3.3.2	5610
Crocking	3.3.2	5651
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave [1]	3.4	Visual
Nonfibrous material	3.6	2611
Shrinkage or stretch:		
Warp	3.7	5550
Filling	3.7	5550
Acidity (pH)	3.8	2811

[1] One deteminration shall be made from each sample unit and the result shall be reported as "pass or fail".

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be expressed in units of 1 yard. All test reports shall contain the individual values utilized in expressing the final results. The sample size (number of sample units) shall be in accordance with the following:

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and up	5

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Levels A and C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP-P-1134.

5.2 Packing. Packing shall be A, B, or C (see 6.2 and 6.2.1).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth covered by this specification is primarily intended for use in the manufacture of clothing items.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type, class, and style required (see 1.2.1).
(c) Color of cloth required (see 3.3 and 6.4).
(d) When colorfastness required is other than specified (see 3.3.2).
(e) When weave required is other than specified (see 3.5).
(f) Width required (see 3.5.1).
(g) When length is other than specified (see 3.9).
(h) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices and alternatives, as indicated in PPP-P-1134, may be exercised in the preparation for delivery of this cloth.

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Note. A suggested but not mandatory dyestuff formulation for Blue 3301 is the use of Vat Blue 43 for the warp yarns and Denivat Everblack-W blended with natural cotton for the filling yarns.

CUSTODIANS:
Navy—Naval Supply Systems Command
Air Force—82

Review interest:
Air Force—85

User interest:
Navy—C.G.

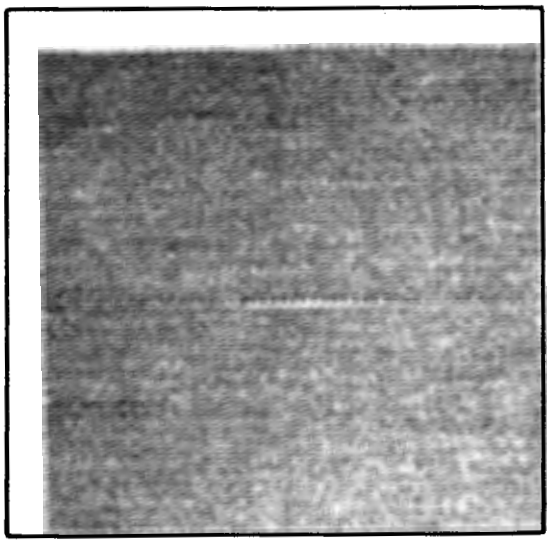
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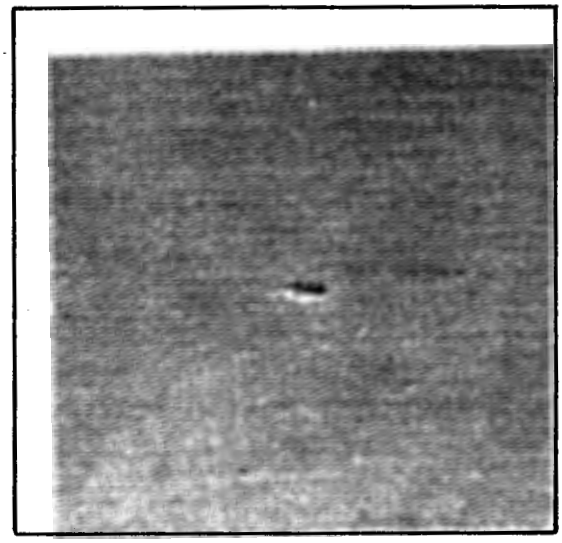
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Figure 1 (Limits of Acceptability)

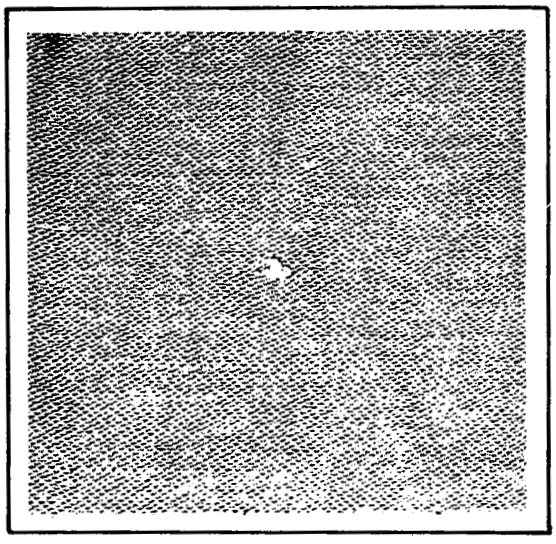
Slub



Slub (Loose, Thick)



Lump or Knot (Fuzz)



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